Tampere University

Nam Từ

Decoding Glioblastoma Through Multiomic Integration of Cell Death Pathways

Bachelor's Thesis
Engineering and Natural Sciences
July 2026

ABSTRACT

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Science and Engineering
Bachelor's Thesis
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The abstract is a concise, self-containing one-page description of the work: what was the problem, what was done, and what are the results. Do not include charts or tables in the abstract. Put the abstract in the primary language of your thesis first and then the translation when that is needed. International students do not need to include an abstract in Finnish.

Keywords: keyword1, keyword2, ...

The originality of this thesis has been checked using the Turnitin Originality service.

PREFACE

"DNA neither cares or knows. DNA just is. And we dance to its music."

— Richard Dawkins

This thesis was written as part of Tampere University's Bachelor's Programme in Science and Engineering and was begun in early May 2026. The work was supervised by Dr. Meenakshisundaram Kandhavelu, who inspired my interest in bioinformatics and provided me the opportunity to develop it further in his Molecular Signaling Lab.

This work also marks an important milestone not only for myself but also for my family.

I would like to thank my friends from various parts of the world for their encouragement, insight, and support throughout the writing process — both online and in-person. Finally, I express my deepest gratitude to my parents, V. and K., whose support made this achievement possible. The completion of this thesis—and the degree as a whole—would not have been possible without any of them.

In Tampere on 18. July 2026,
Nam Từ

Contents

Glossary

GBM	Glioblastoma multiforme
$\mathbf{M}$	Upright bold capital letters denote matrices.
Multiomics	Research that combines biotechnology ending with the “-omics” suffix such transcriptomics, proteomics, genomics, etc.
s	Lower-case italic letters denote scalars.
TUNI	Tampere University
TUT	Tampere University of Technology
$\mathbf{v}$	Bold upright lower-case letters denote vectors.

USE OF ARTIFICIAL INTELLIGENCE IN THIS WORK

Artificial intelligence (AI) has been used in generating this work:

☒ Yes
☐ No

I hereby declare, that the AI-based applications used in generating this work are as follows:

Application	Version
ChatGPT	GPT-5.5

PURPOSE OF THE USE OF AI

Explain here *in detail*, for which purpose and how AI was utilized in writing this thesis.

ChatGPT was used mainly for debugging code during analysis of multiomics.

PARTS OF THIS WORK, WHERE AI WAS USED

List here all chapters, sections, subsections, tables, figures and so forth, that were generated by an AI, or that an AI had a hand in generating.

ACKNOWLEDGEMENT OF RISKS

I hereby acknowledge, that as the author of this work, I am fully responsible for the contents presented in this thesis. This includes the parts that were generated by an AI, in part or in their entirety. I therefore also acknowledge my responsibility in the case, where use of AI has resulted in ethical guidelines being breached.

1 INTRODUCTION

Glioblastoma multiforme (GBM) is the most aggressive primary brain tumour in adults and remains associated with poor patient survival despite advances in treatment [1]. One of the major challenges in understanding GBM is its extensive molecular heterogeneity, with tumours exhibiting distinct molecular subtypes characterised by unique transcriptional programmes and biological behaviours. These differences influence tumour progression, treatment response, and patient prognosis, making subtype-specific molecular analysis an important area of research.

Advances in high-throughput transcriptomic technologies, together with AI-driven analytical methods, have enabled increasingly comprehensive investigations of these subtype-specific molecular profiles [2]. Interpreting these data within their biological context is therefore essential for identifying dysregulated pathways and potential therapeutic targets.

This thesis investigates transcriptomic alterations across GBM molecular subtypes through differential expression analysis relative to healthy brain tissue. Functional interpretation of the resulting differentially expressed genes and proteins is performed using Gene Ontology enrichment analysis, followed by integration with genes involved in regulated cell death pathways, including apoptosis, autophagy, ferroptosis, necroptosis, and pyroptosis. By combining differential expression with functional enrichment and pathway-based analyses, this work aims to provide insight into subtype-specific biological processes and the potential roles of regulated cell death mechanisms in GBM.

Chapter 2 presents the relevant knowledge regarding the thesis's topic including glioblastoma and multiomics of glioblastoma. Chapter 3 shows the processing of expression data, enrichment of the resulting significant gene set, and integration of cell death pathway-related genes. In-depth results and discussion are presented in Chapter 4.

2 Theoretical Background

2.1 Glioblastoma biology

Glioblastoma multiforme is one of the most aggressive and most common form of brain tumors, with an extremely poor prognosis of a median survival period of 12-15 months [1], [2]. GBM possesses an extreme heterogeneity in its structure which results in its resistance to conventional therapies and high recurrence rate [2].

Advanced sequencing technologies emphasizes the need for therapies to target both molecular drivers of the tumors and their arised microenvironment; one of the promising path to battle GBM involves precision medicine, leveraging computational methods for targeted therapy selection and resistance prediction [2].

2.2 Transcriptomics and Differential Expression Analysis

Transcriptomics technologies obtains information on how genes are expressed an organism's DNA by looking through all the RNA transcripts produced from the DNA [3], [4]. According to Lowe et al. [4], there are two main ways of approaching the reading of RNA, including (1) quantifying predetermined sequences with microarrays, and (2) using RNA sequencing (RNA-seq) which also identifies sequences that were not predetermined via high-throughput sequencing. The Clinical Proteomic Tumor Analysis Consortium (CPTAC), among with other public databases, has deposited a large amount of transcriptomic profiling data for numerous tumor types including that of glioblastoma [5]. The thesis utilizes CPTAC's GBM Discovery Study data.

Comparitive profiling is the process of comparing expression levels of one sample against another (e.g. a diseased sample against a healthy sample) to identify and understand molecular causes of biological processes [6]. It is the main form of computation done in this thesis.

2.3 Gene Ontology

The Gene Ontology (GO) database provides structured and standardized representation of biological activities according to gene products, organized into three main aspects: (1) Molecular Functions (MF), (2) Cellular Components (CC), and (3) Biological Processes (BP) [7].

Gene Ontology enrichment analysis is commonly performed following differential expression analysis to determine whether particular GO terms — items belonging to each of the aforementioned aspects — are statistically overrepresented within a set of differentially expressed genes compared with a suitable background gene set instead of inspecting individual genes in isolation [8], [9]. In the context of glioblastoma, GO enrichment provides insight into biological mechanisms associated with each molecular subtype.

Although all three GO ontologies provide complementary biological information, this thesis focuses primarily on the Biological Process ontology, as it describes coordinated cellular pathways and processes that can subsequently be related to regulated cell death signalling pathways.

2.4 REGULATED CELL DEATH SIGNALING PATHWAYS

One of the main processes cells experience — especially after they have served their purpose — is programmed cell death, occuring via a complex cascade of molecular signaling. It keeps the bodily functions in check, and problems arise when this process is hindered or accelerated unexpectedly [10], [11]. The core types of programmed cell deaths are [11]:

- **Apoptosis**: Regulated through intrinsic (mitochondrial) and extrinsic (death receptor) signalling pathways.
- **Autophagy**: Regulated through signalling pathways controlling lysosomal degradation and cellular recycling.
- **Necroptosis**: Initiated when death receptor signalling occurs while caspase-8 activity is inhibited, preventing apoptosis.
- **Pyroptosis**: Inflammatory programmed cell death driven by inflammasome signalling.
- **Ferroptosis**: Regulated by signalling pathways governing iron metabolism, lipid peroxidation, and cellular redox balance.

Each programmed cell death mode is regulated by a characteristic network of genes and proteins [11]. Consequently, differential expression of genes participating in these pathways can provide insight into alterations in the underlying signaling mechanisms. By integrating differentially expressed genes with Gene Ontology enrichment and curated cell death pathway gene sets from dedicated pathway databases such as KEGG or Reactome, it becomes possible to identify which regulated cell death pathways are most strongly associated with each GBM molecular subtype.

3 Research methodology

3.1 Data retrieval and preparation

Both transcriptome and proteome data was obtained from CPTAC GBM Discovery Study dataset, given as already-cleaned-and-normalized (but not yet $\log_2$-transformed) expression values according to their originating gene. The steps for this section were performed twice: once for the transcriptome data, another for the proteome data.

Transcriptomic and proteomic expression values typically span several orders of magnitude and exhibit right-skewed distributions [12]. Therefore, a common practice when working with expression matrices, which has been done to the current data, is to apply a $\log_2$-transformation to each of the value [12], [13]. An example expression value matrix ready for calculation is given in Table 3.1. From this point onwards, "$\log_2$ expression" is now referred to as "expression".

Table 3.1. Example expression matrix

Gene	Sample 1	Sample 2	Sample 3	...
A1BG	18.833	18.647	18.958	...
SPC25	13.475	13.469	13.786	...
...	...	...	...	...

3.2 Performing Differential Analysis

3.2.1 Calculation & Testings

In the CPTAC Discovery Study, healthy brain tissue samples are provided as the baseline for comparison for the remaining tumor samples. As described in 2.2, comparing the gene expressions of the tumors against healthy samples can give insights (or, at least, a base for further comparative computations) into what is different.

The samples were split into three major subtypes according to the metadata: classical, proneural, mesenchymal. For each subtype, the $\log_2$-fold-change (log2FC) for a specific gene becomes:

$$\log_2 FC = \frac{\sum(\text{expression in subtype})}{n_{\text{subtype}}} - \frac{\sum(\text{expression in healthy sample})}{n_{\text{healthy}}} \tag{3.1}$$

which means the difference between the average expression of a gene in the subtype and the average expression of the same gene in the healthy samples.

At the same time, the Welch's t-test for independence is applied to obtain the statistical significance of the difference between the expression of the tumors and healthy samples for a specific gene in the form of the obtained p-value [14]. The Welch's t-test was chosen over the standard Student's t-test due to the high chance of the variances to be different [14]. The analysis was implemented in Python using SciPy's `ttest_ind` function with `equal_var=False`.

To account for inflated false-positive count resulting from multiple testings, the p-values are corrected with the Benjamini-Hochberg method for False Discovery Rate (FDR) [15]. This is achieved in Python with the `statsmodels` library using the `multipletests` function and `method=fdr_bh` parameter.

With the obtained p-value and log2FC obtained for each gene, the genes are classified whether they are considered statistically and quantitatively significant with a chosen threshold for p-value and absolute value of log2FC, which are 0.05 and 1 respectively for this thesis. Consequently, genes with $p\text{-value} < 0.05$ and $\log_2 FC > 1$ are categorized as "Significantly Upregulated", and genes with $p\text{-value} < 0.05$ and $\log_2 FC < -1$ are "Significantly Downregulated". From this point onwards, the categories are simply referred to as "upregulated" and "downregulated".

This step concludes the preparation for a "solid base" of results for further comparative computations, which should include: (1) gene name and its corresponding (2) transcriptome log2FC, (3) transcriptome adjusted p-value, (4) proteome log2FC, (5) proteome adjusted p-value.

3.2.2 Exploratory Plotting

With the results, we can plot a variety of plots to give insight into the overall structure of the expressions. For instance, the volcano plot: log2FC for the x-axis, $-\log_{10}(p\text{-value})$ for the y-axis. Additionally, the thresholds can be added to the plot to show the categorically significant portion of the distribution. For example, the volcano plots for the expression values of both transcriptome and proteome for the proneural subtype are shown side-by-side in Figure 3.1.

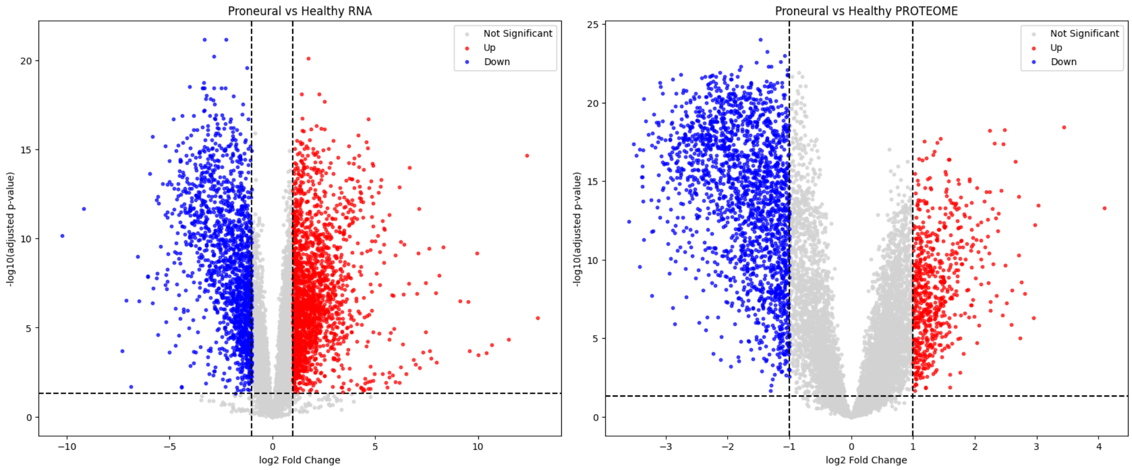

Figure 3.1. Example volcano plots

Another possible plot to see the structure of the data include plotting log2FC of the transcriptome against that of the proteome, with this we can calculate the correlation between the two. Additionally, the distribution of log2FC values can be inspected in a density plot. Example for both, produced from the processed CPTAC expression data, are shown in Figure 3.2.

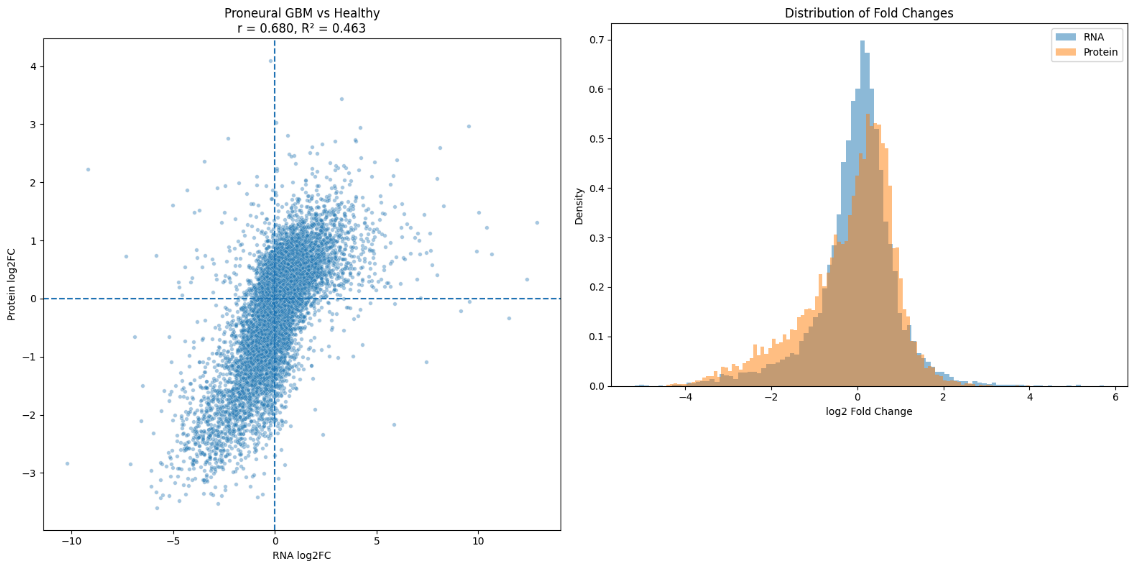

Figure 3.2. Example log2FC comparison plots

The resulting $R^2 = 0.463$ from the correlation of log2FCs lines up quite closely with previous major studies, where only about 40% of the variation of one can be explained by the other [16].

3.3 GENE ONTOLOGY ENRICHMENT AND CELL DEATH PATHWAY INTEGRATION

From the set of significant genes for individual subtypes, gene set enrichment analysis was applied to identify significant changes in cellular aspects that would otherwise go unnoticed if examination was done on an individual gene basis [17]. This was achieved by using the `GSEApy` library, and using the library's predefined gene set Gene Ontology (GO) libraries `GO_Biological_Process_2023`,`GO_Cellular_Component_2023`, and `GO_Molecular_Function_2023` to identify biological processes, cellular components, and molecular functions significantly affected by the genes. One can choose the gene set fed into `GSEApy` to include only upregulated genes, downregulated genes, or both. For explatory purposes, all three options can be done. For this thesis, however, including both was chosen to capture as much of the underlying interactions as possible for further analysis. Results are presented in section 4.1. Additionally, the set of genes involved in the biological processes affected were collected for further integration with cell death pathways.

Sets of genes in the five major modes of cell death mentioned in 2.4 were also obtained with `GSEApy` from `KEGG_2021_Human` and `Reactome_2022` pathway databases.

4 RESULTS AND DISCUSSION

4.1 GENE ONTOLOGY ENRICHMENT

4.2 CELL DEATH PATHWAY INTERGRATION

5 INTRODUCTION

This document template conforms to the Guide to Writing a Thesis in Tampere University [18], [19]. A thesis typically includes at least the following parts:

- Title page
- Abstract in English (and in Finnish)
- Preface
- List of abbreviations and symbols
- (Lists of figures and tables)
- Contents
- Introduction
- Theoretical background
- Research methodology and material
- Results and analysis (possibly in separate chapters)
- Conclusions
- References
- (Appendices)

Each of these is written as a new chapter, in the files in the folders `frontmatter/`, `mainmatter/` and `appendices`, which are then included into the `index.typ` files in said folders using the Typst directive `#include`. Read this document template and its comments carefully. The titles of Chapters from 1 to 8 are provided as examples only. You should use more descriptive ones. The title page and abstract headers are created by inserting the relevant information into the file `metadata.typ`. The table of contents lists all the numbered headings after it, but not the preceding ones.

Introduction outlines the purpose and objectives of the presented research. The background information, utilized methods and source material are presented next at a level that is necessary to understand the rest of the text. Then comes the discussion regarding the achieved results, their significance, error sources, deviations from the expected results, and the reliability of your research. The conclusions form the most important chapter. It does not repeat the details already presented, but summarizes them and analyzes their consequences. A list of references enables your reader to find the cited sources.

An introduction ends in a paragraph, that describes what each following chapter contains. This document is structured as follows: Chapter 6 discusses briefly the basics of writing and presentation style regarding the text, figures, tables and mathematical notations. Chapter 7 summarizes the referencing basics. Chapter 8 presents a discussion on the discoveries made during this study.

6 Writing practices

Effective written communication requires both sound content and clear style. Keep the layout of your thesis neat and pay attention to your writing style.

6.1 Text

Do not worry about the layout of the text, this template takes care of it already. Brief basics of writing style:

- Always think of your reader when you are writing and proceed logically from general to specific.
- Highlight your key points, for example, by discussing them in a separate section or subsection, or presenting them in a table or figure. Use *italics* (`_text_`) for emphasis, but don't overdo it.
- Avoid long sentences and complicated statements. A full stop is the best way to end a sentence.
- Use active verbs to make a dynamic impression but avoid the first person pronoun I, except in your preface.
- Avoid jargon and wordiness. Use established terminology and neutral language.
- The minimum length of sections and subsections is two paragraphs, and you need to consider the balance of chapters. Paragraphs must always consist of more than one sentence.
- Do not use more than three levels of numbered headings, such as $4.4.2$. However, keep in mind that 3 different levels of sections is already a bit much.
- Do not use too many abbreviations. Use capital and small letters consistently. You might wish to define new commands in the `preamble.typ` file, for words you do not wish to misspell. For example, the `preamble.typ` file defines the command `#eeg`, which is typeset as electroencephalography.

Your text will also look better if you do not extensively rely on lists such as the above one. Ideally, a list should not end a section either, which is why this paragraph is here.

6.2 Images

You must refer to all the images in the body text. The reference should preferably appear on the same page as the actual image or before it. Images and tables must be numbered consistently and primarily placed at the top of the page, but you are free to decide where they fit best.

6.2.1 IMAGES IN TYPST

Typst takes care of the numbering, if you specify a unique label `<label-name>` right after the a image or a table. Cross-referencing is done by prefixing the identifier inside the angle brackets with a `@` symbol: `@label-name`. For example, the code

```
#figure(
    image(
        "../images/tau-logo-fin-eng.svg",
        alt: "Tampere University logo with a human face formed from geometric shapes on the left and the university name on the right.",
        width: 80%
    ),
    caption: [Tampere University logo.],
) <tau-logo>
```

is rendered as seen in Figure 6.1.

Tampereen yliopisto
Tampere University

Figure 6.1. Tampere University logo. [20]

To showcase how figures with subfigures are generated, Figure 6.2 displays a few different gradient or color progression shapes. Other colormaps than the `rainbow` seen here are also available. The subfigures are generated by calling the function `tauthesis.subfigure` inside of a figure of type `image`.

(a) (b) (c)

Figure 6.2. A display of a few different color gradient shapes available in Typst. These include the linear gradient in Figure 6.2.a, the radial gradient in Figure 6.2.b and the conic gradient in Figure 6.2.c.

Never start or end a chapter with a image, table, equation or list. The caption is placed under an image. The contents of all images must be explained in the text body, so that the readers know what they are supposed to notice. Images generated by analysis software usually need further editing. The figures should be in the same language as other text. The recommended font size is the same as that of the body text, $11\,\mathrm{pt}$. All images must be readable, even if your thesis is printed in greyscale. Whenever possible, use images in vector formats such as `.svg`, as they can be scaled without loss of quality. Also note that as of Typst version 0.14.0, including PDF files as images should work natively. However, see the below accessibility considerations for why this might not work for your thesis.

6.2.2 Accessibility of images

All images produced via the `image` [21] function (or otherwise) *must* contain so-called *alternative texts* [22], provided via the named argument `alt` . These **must** be provided for your thesis to remain accessible. Even if an `image` is inside of a captioned `figure` [23], an alternative text is still needed. The alternative text also needs to be more descriptive than the figure caption, because an alternative text is the only indication of what goes on in an image for people relying on assistive technologies. You can see an example of these principles being applied in Figure 6.1.

Also note that since most PDF files produced programmatically do not contain tags, and Typst has no way of automatically deducing what the tags should have been like, including PDF images does not work with the accessible PDF standards such as PDF/UA-1, PDF/A-3a and others. You should therefore prefer SVG images when inserting vector graphics into your thesis.

6.3 Tables

Tables are convenient for presenting information in a concise way, especially numerical data. Tables have numbered captions, see Table 6.1 for an example. The caption is placed on the same page but above the table, unlike the captions that accompany images. You must refer to all the tables in the body text. In addition, you must discuss the content of any tables in the body text to ensure that readers understand their relevance.

Mark the titles of the columns and units clearly. You can use underscores `_emph_` to highlight the titles, if necessary. The order of the columns and rows must be carefully considered. Ideally, all cells should *not* be surrounded with a border, as it may make your table harder to read. The numbers should be aligned at the decimal point for easy comparison. You should preferably use SI units, established prefixes and rewrite large numbers so that the power of ten should be placed in the title of the column instead of each row, if possible. More suggestions can be found in [24].

6.3.1 TABLES IN TYPST

To actually draw a table that can be referenced, we issue a `#figure` call with a `table` as its argument:

```
#import table: cell, header, hline, vline
#[
    #show table.cell.where(y: 0): strong
    #figure(
        table(
            columns: 5,
            stroke: none,
            align: center + horizon,
            table.header(
                [Substance],
                table.vline(),
                [Pressure (#unit(pascal))],
                table.vline(),
                [Pressure (#unit(bar))],
                table.vline(),
                [Pressure (#unit(mmHg))],
                table.vline(),
                [Temprature (#celsius)]
            ),
            table.hline(),
            [Mercury],
            num(1.0),
            num(0.0001),
            num(0.0075),
            num(41.85),
            [Tungsten],
            num(1.0),
            num(0.0001),
            num(0.0075),
            num(3203),
        ),
        caption: [Vapour pressures of two metals. @wikipedia-vapor-pressure],
    ) <tab-evaporation-conditions>
]
```

The added `#[...]` around the whole figure restricts the scope of the show rule that boldens the header to this table. This results in Table 6.1 being displayed.

Table 6.1. Vapour pressures of two metals. [25]

Substance	**Pressure** (Pa)	**Pressure** (bar)	**Pressure** (mmHg)	**Temprature** (°C)
Mercury	1	0.000 1	0.007 5	41.85
Tungsten	1	0.000 1	0.007 5	3 203

A table should not end a section. That is why after every table we state some observations about the table. Here we notice that mercury and tungsten have identical vapour pressures, but at very different tempratures.

6.3.2 Accessibility of tables

Tables do not need alternative texts, not even if you export your document using the PDF/UA-1 standard. Screen readers will read the table in the order in which your table cells were given in Typst source code. It is also important to properly use the `table.header` and `table.footer` for accessibility tags to be applied properly.

The example table shows how to typeset a table in the usual *row-major* order, where the displayed order in the code is the same as it is in the final PDF file. There are situations, such as when a table takes too much space horizontally, where you might wish to *transpose* a table in the PDF output, so the output is in *column-major* order. Typst does not currently support this in tables that aim to be semantically correct, as table headers and footers constructed with `table.{header,footer}` are always typeset horizontally, and `table.cell` show rules currently (as of 2025-10-09) do not have access to the `x` and `y` positions of each cell.

Also note that accessible tables should be *regular* [26]: every row of a table should contain an equal number of columns or cells. In other words, table cells spanning multiple columns or rows are heavily discouraged. For the more mathematically oriented, it could be said that tables should only be used to represent *relations*. Table 6.1 is an example of such a regular table.

If your table is getting very wide, consider simply scaling it to the page width using the `scale` function. Note however that this will make the table contents tiny, and less accessible to people who do not see well, but still do not rely on screen readers. Rotating tables by $90°$ using the `rotate` function is also possible.

6.4 Mathematical notations and equations

Numbers are generally written using numerals for the sake of clarity, for example "6 stages" rather than "six stages", which is nevertheless strongly preferred to "a couple of stages". You should also use a thousand separator, i.e. instead of 55700125 write $55\,700\,125$. Never omit the leading zero in decimals. A comma is used as a decimal separator in the Finnish language and a period in the English language. You might wish to use the Typst library **zero** for typesetting numbers, even if the file `preamble.typ` does provide a rudimentary function `num` for grouping number digits.

Like numbers, it is advisable to abbreviate units of measurement. There is a space between the number and the unit, but you must keep them on the same line. The space is somewhat shorter than a word space, see 1 μm and $1\,\mathrm{\mu m}$ for comparison. It is better to compile a table or graph than include a great deal of numerical values in the body text. Use precise language and put numbers on a scale (small, fast, expensive).

Use generally known and well defined concepts and standard conventions and symbols for representing them. New concepts should be defined when they appear in the text for the first time. Upper case and lower case letters mean different things in symbols and units of measurement. Do not use the same symbol to mean different things, but note that different styles applied over the same symbol are accepted: if $\mathbf{x}$ is a position vector, then the scalar $x = \|\mathbf{x}\|$ could be used to denote its norm. If the position is a random variable, then one could use yet another style such as fraktur form $\mathfrak{x}$ of $\mathbf{x}$ to eccentuate this fact.

6.4.1 Mathematics in Typst

Strings of mathematical symbols such as $\Theta(n^2)$ are typeset in Typst using the math mode, between dollar signs: `$math$` for inline mathematics or

```
$
    math
$
```

with whitespace surrounding the contents for display math. Simple formulas may be displayed within the body of the text without numbering. As an example of a highlighted formula, the Fundamental theorem of calculus can be written in the following way:

$$\int_{[a,b]} f(x)\,\mathrm{d}x = F(b) - F(a)\,. \tag{6.1}$$

Here f is the *integrand*, F its *anti-derivative* and $[a, b]$ the integration interval. Please note that all the variables must be defined at the point of their first appearance. The formulae are shown in a different typeface on purpose and the symbols are almost always italicised. Vectors can be written in slanted or upright boldface, or with arrows:

$$\begin{gathered} \boldsymbol{v} = \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} \\ \mathbf{v} = \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} \\ \vec{v} = \frac{\mathrm{d}\vec{x}}{\mathrm{d}t} \end{gathered} \tag{6.2}$$

The most important thing is to be consistent throughout your thesis. Do not mix and match notations, as it will confuse the reader. See the file `preamble.typ` for an example of how you might define a `#vector` command for typesetting vectors consistently.

Defining a command also allows you to easily switch notations globally, if you are not satisfied with your current choice.

Units are written in an upright font with a normal weight, to distinquish them from variables:

$$\|\mathbf{f}\| = m\|\mathbf{a}\| = 10\,\mathrm{kg} \cdot 9.81\,\frac{\mathrm{m}}{\mathrm{s}^2} = 98.1\,\mathrm{N}\,. \tag{6.3}$$

Here $\mathbf{f}$ is a force vector, m is the mass of the object experiencing the force, and $\mathbf{a}$ is its acceleration vector. Again, see the `preamble.typ` file for how you might define commands for units.

Mathematical formulae are numbered, if they are written on separate lines and referred to in the main body of the text. The number is usually put in parenthesis and right aligned, see (6.1) for an example. If you wish to leave out the number from a highlighted equation, wrap it into the function `math.equation` with the named arguments `numbering: none` and `block: true`:

```
#math.equation(
    numbering: none,
    block: true,
    alt: "force = mass acceleration .",
    $force = mass acceleration thin .$
)
```

This results in the unnumbered block-level equation

$$\mathbf{f} = m\mathbf{a}\,.$$

Include any punctuation (commas, periods, …) surrounding an equation in the equations themselves, as is again shown in (6.1). The command `thin`, which in mathematics mode is a shorthand for `sym.space.thin`, can be used to insert just a little extra space between elements such as the math itself and punctuation, which is a good readability-increasing practice. The added visual space is not relevant for the alternative text of the equation, though, so `thin` should be removed from it, as is shown above.

Do not start a sentence with a mathematical symbol but add some word, such as the name or type of the symbol in front of it. Variables, such as x and y, are generally presented in italics, whereas elementary functions, special functions and multi-letter operators are not:

$$\sin(2x + y) \quad \text{or} \quad \lim_{x \to -1} \frac{x^2 - 1}{x + 1} = -2\,. \tag{6.4}$$

As a rule of thumb, it is best to rely on the automated formatting of an equation editor, such as the one provided by Typst.

Vectors and matrices are written with the `vec` and `mat` functions:

$$\mathbf{A}\mathbf{x} = \begin{bmatrix} \mathbf{A}_{1,1} & \mathbf{A}_{1,2} \\ \mathbf{A}_{2,1} & \mathbf{A}_{2,2} \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \end{bmatrix} = \mathbf{b}\,. \tag{6.5}$$

See the Typst documentation on mathematics [27] for examples of how different symbols are written.

6.4.2 Accessibility of mathematics

If you have been reading the source code of the PDF file of this example template, you might have noticed that in addition to the math mode dollar signs `$math$` and

```
$
    math
$
```

the template has explicitly been using the Typst function `math.equation` [28] to include mathematics into a document. This is because the function allows one to provide an *alternative text* for each equation, as is required by the PDF accessibility standards. This is done via the alternative text argument `alt` of the function:

```
#math.equation(
    alt: "Alternative text here",
    $equation here$
)
```

You might have noticed that the alternative texts in the example mainly use the Typst source code as the alternative text representation. **Note that this is only acceptable, if your equation source code carries the semantics of the displayed equation with it.** Note the difference between the *completely unacceptable*

```
#math.equation(
    alt: "f = m a",
    $vector(f) = m vector(a)$
)
```

the slightly better option

```
#math.equation(
    alt: "vector(f) = m vector(a)",
    $vector(f) = m vector(a)$
)
```

and finally the most descriptive one, where the commands `force`, `mass` and `acceleration` have been defined in the file `preamble.typ`:

```typst
#math.equation(
    alt: "force = mass acceleration",
    $force = mass acceleration$
)
```

Remember that in equation (6.3) the text styles (bold versus normal weight) were used to differentiate between scalars and vectors. Force and acceleration are implicitly vectors, and mass is implicitly a scalar. Having the alternative text mention what the actual quantities are instead of using just letters makes the intention clearer. It also allows you to copy and paste the source code into the alternative text without any modifications.

You might also wish to use the function `altTextFn` provided in `preamble.typ` to clean up extra whitespace from your alt texts, if you wish to write them on multiple lines. Indentation and added space are not automatically trimmed from alt texts by Typst.

6.5 PROGRAMS AND ALGORITHMS

Codes and algorithms are written using a monospaced font. If the length of the code or algorithm is less than 10 lines and you do not refer to it later on in the text, you can present it similarly to formulas. If the code is longer but shorter than a page, you present like a figure titled “Listing” or “Algorithm”. In Typst, code blocks are presented in backticks:

````
```typst
#figure(
    raw(
        read("../code/square.jl"),
        lang: "julia",
        block : true
    ),
    caption: [
        The square of
        #math.equation(
            alt: "x in real numbers",
            $x in bb(R)$
        )
        and
        #math.equation(
            alt: "x in complex numbers",
            $x in bb(C)$
        ),
        written in Julia @Bezanson2017Julia.
    ],
) <program-example>
```
````

The above results in what is seen in Listing 6.1.

```
#figure(
    raw(
        read("../code/square.jl"),
        lang: "julia",
        block : true
    ),
    caption: [
        The square of
        #math.equation(
            alt: "x in real numbers",
            $x in bb(R)$
        )
        and
        #math.equation(
            alt: "x in complex numbers",
            $x in bb(C)$
        ),
        written in Julia @Bezanson2017Julia.
    ],
) <program-example>
```

Listing 6.1. A code block.

Removing the backticks would actually load code from the file `code/square.jl` to be displayed in the numbered Listing 6.2. The listing demonstrates how multiple dispatch allows you to define methods for functions with different argument types in the Julia programming language.

```
"""
  out = square(x::Real)

Computes the square of an x ∈ ℝ.
"""
function square(x::Real)
  x * x
end

"""
  out = square(x::Complex)

Computes the square of an x ∈ ℂ.
"""
function square(x::Complex)
  conj(x) * x
end
```

Listing 6.2. The square of $x \in \mathbb{R}$ and $x \in \mathbb{C}$, written in Julia [29].

Again, listings should not end sections. You should always describe what is going on in the displayed code after presenting a listing.

6.6 THEOREM BLOCKS

Example 6.1 and Esimerkki 6.2 show examples of theorem blocks:

Example 6.1 (Pythagorean theorem). For sides a, b and c of a right triangle, where c is the *hypotenuse*, the following equality holds: $a^2 + b^2 = c^2$.

Proof. Proof goes here. ∎

Esimerkki 6.2 (Pythagoraan lause). Suorakulmaisen kolmion sivut a, b ja c, missä c on *hypotenuusa*, toteuttavat yhtälön $a^2 + b^2 = c^2$.

Todistus. Todistus tulee tähän. ∎

As one can see, these follow the same counter. The theorem box has been typeset with the command seen in Listing 6.3.

```
#tauthesis.example(
    title: [Pythagorean theorem],
)[
    For sides
    #math.equation(
        alt: "a",
        $a$
    ),
    #math.equation(
        alt: "b",
        $b$
    )
    and
    #math.equation(
        alt: "c",
        $c$
    ),
    where
    #math.equation(
        alt: "c",
        $c$
    )
    is the _hypotenuse_, the following equality holds:
    #math.equation(
        alt: altTextFn("
            a^2 + b^2 = c^2 .
        "),
        $
            a^2 + b^2 = c^2 thin .
        $
    )
] <pythagorean-theorem>
```

Listing 6.3. Example code for displaying a theorem block.

The reference label is placed after the call to `tauthesis.example` via the `<pythagorean-theorem>` syntax. The module `tauthesis` defines the following theorem box functions:

- `definition`,
- `theorem`,
- `lemma`,
- `corollary` and
- `example`.

Their Finnish equivalents

- `määritelmä`,
- `lause`,
- `apulause`,

- `seurauslause` and
- `esimerkki`

are also defined. These all follow the same counter, and can all be referenced by suffixing the equivalent function call with a tag.

Try not to overuse these, however. Academic theses should *not* be written in the *Landau style*, where the entire text consists of definitions after definitions and theorems after theorems. There should always be a full paragraph between two theorems or other objects, such as figures and tables.

6.7 ACCESSIBILITY CONSIDERATIONS

In accordance with the European Union directive 2016/2102 [30], Finnish law [31] requires that all electronic publications of public sector bodies be made *accessible*, as long as doing so would not require an infeasible amount of work. This also applies to academic theses published by universities, including yours [32]. For this reason you should be aware of the requirements related to accessible electronic documents. Some of these considerations are described below. These are also discussed in the Typst accessibility guide [22].

Use of language: Use simple and descriptive language. Avoid overtly long sentences. Any truly essential information should be delivered in words and not by visual means.

Figures: As per the point above, any figures in the document should have some kind of a textual alternative available. Ideally so called *alternative texts* [33], [34], [35] (which are not the same thing as figure captions) should be included into a document. However, if this is not feasible due to current technological limitations, one should at least try to describe what is happening in each image or table in the body text as well as they can. The Typst functions `image` [21] and `figure` [23] accept the named argument `alt`, which allows you to provide an alternative text for an included image.

Document metadata: This should be as complete as possible. However, the most essential metadata fields are the *title* of the work, and the *main language* the work was written in. This is relevant from the point of view of assistive software like screen readers, that visually impaired consumers of a document might be relying on. Also, resources such as the fonts intended to be used with a document should be embedded into the document itself.

Mathematics: As professor Antti Valmari previously of Tampere University of Technology fame once said in a conversation with one of the maintainers of this template, “Mathematicians have an inexplicable urge to put things on top of each other.”. From the point of view of accessibility of PDF files, this is a terrible practice, so long as the document format does not include either structural information related to the semantics of the visual presentation within it, or alternative texts that screen reader sofware can access. This is because classically PDF is a format where graphical elements are essentially drawn on a canvas with x- and y-coordinates, which becomes apparent if one tries to copy and paste the following fractions from the output PDF file of this template:

$$a \div b\,, \quad a/b\,, \quad \frac{a}{b} \quad \text{and} \quad ab^{-1}\,. \tag{6.6}$$

Depending on your PDF reader implementation, it might be that with the vertically stacked fraction $\frac{a}{b}$, the numerator a ends up on the previous line, since it is vertically above the line where the fraction itself resides, and reading order is based on the coordinates (elements higher and more to the left come first). Alternatively, it might be that the stacked fraction is handled as a whole, but since the horizontal fraction line separating the numerator a and denominator b is implemented as a graphical element, which does not have a textual representation in the PDF model, the line will be lost, and you get something like $a\,b$ as the output. This is semantically the complete opposite (or inverse) of the indended meaning of the fraction denominator!

The denominator $1 \div b$ expressed with a negative exponent b^{-1} is just as problematic, since upper and lower indices also don't generally have a proper textual representation in Unicode. When copying and pasting such an expression, you might end up with something like $ab - 1$ in the textual output. This also completely changes the semantics of the output expression.

As of October 2025, neither Typst or LaTeX currently support automated creation of so-called *well-tagged PDF files* [36], which are documents carrying structural information similar to HTML tags. Typst will start introducing support for automatically producing tagged PDF files starting from version 0.14.0 [37], [38]. LaTeX has some experimental support for producing tagged PDFs starting from TeX Live 2024 [39], but it requires intervention from the user of the language, such as manually inserting LaTeX commands to generate said tags. Another issue with well-tagged PDFs is that many PDF readers still do not support the standard. While we wait for the related technology such as the embedding of MathML [36], [40] into PDF files to catch up, there are some simple things you can do to make your thesis more accessible.

6.7.1 ADD ALTERNATIVE TEXTS OR OTHER STRUCTURE TO YOUR EQUATIONS

The importance of this was explained in Section 6.4.2. Typst version 0.14.0 supports adding alternative texts to equations through the function `math.equation` [28] named argument `alt`. In future versions on Typst, it is also intended to support the PDF/UA-2 standard, which would allow automatically embedding structural information in the form of MathML into a PDF document. This would remove the need for alternative texts in the case of equations.

Another means of keeping text accessible is to keep the flow of the text as horizontal as possible, which might allow text extraction programs such as `pdftotext` [41], which also operate on the coordinate system basis, to extract the text in a correct reading order. Paul Dirac did a similar thing with his bra–ket notation for linear algebra products. This enabled him to write mathematics with a normal chevron-equipped typewriter, without access to special symbols such as greek letters, upper and lower indices and integrals. Especially, it simplified writing long subindex labels related to linear operators such as integrals [42]:

$$\langle i \mid j \rangle \coloneqq \int \varphi_i * \psi_j \, \mathrm{d}x \, . \tag{6.7}$$

Here φ and ψ denote wave functions, and the left-associative unary operator $a*$ denotes the complex conjugate of its argument a. However, inventing your own notations should be done sparingly, since this is against the guideline of sticking to well-established notations. Inventing custom notations also carries with it the burden of having to explain yourself more.

6.7.2 CONVERT YOUR FINAL SUBMISSION TO PDF/UA-1 FORMAT

If you are using Typst version ≥ 0.14.0 locally via a command line, it is possible to produce a thesis into a tagged PDF/UA-1 format via the command

```
typst compile --pdf-standard ua-1 main.typ
```

If the file was generated correctly according to the PDF/UA-1 standard, it should contain accessibility tags. There are not many free tools that can actually visualize these tags for non-visually-impaired people, but the friendly people of the LaTeX Project have an online tool showtags [43] that you might use for this purpose. For the Microsoft Windows® platform, there also exists the tool PAC [44] or *PDF Accessibility Checker*. The PDF Association has also published a list of other tools that might be used to check PDF files for accessibility [45]. Before using either of these, run your thesis through the PDF validator veraPDF [46].

The standard also allows you to embed arbitrary files into your PDF file, using Typst's PDF module function `pdf.attach` [47], and adds accessibility tags to the document. As discussed below, Typst compilers before version 0.14.0 cannot yet produce truly accessible documents, so including the source code of your thesis chapters as plain text provides an alternative means of accessing your thesis. This template has tried to demonstrate how this can be performed at the beginning of each front- and main matter file, and in the example appendices. If you are using the Typst Web application, simply choose PDF/UA-1 as the output format in the file export menu.

Note: if you are unable to produce an accessible version of your thesis according to these PDF accessibility checkers, you should attach the source code of the thesis in a sensible reading order to the Trepo submission! This template shows examples of embedding the source code into the PDF file itself, but note that a separate ZIP archive is the best format for the source code submission. Visually impaired people might not be able to extract the attachments from a PDF file, but a separate ZIP archive can be opened easily on any system.

6.7.3 Compilation dissertations and PDF attachments

If you are writing an accessible compilation dissertation and wish to include scientific articles as attachments to your document[1], you might need to do it in post production. Typst can include PDF files via the `image` function, but most publishers at this point do not care about accessibility that much, and therefore the PDFs downloaded from their websites often do not contain accessibility tags.

The PDF renderer in the Typst compiler cannot handle this lack of accessibilty tags automatically, as there is practically no means of automatically deducing what the tag structure should have been in such a document. Therefore Typst does not support exporting documents in accessible PDF formats, if you attempt to import PDF images into your thesis. However, see Section 7.3 for how you might still do this, to at least preview what your final product might look like.

Note that if you published your work via *open access publishing channels* [48] with a permissive license [49], you can create a *derivative work* of your publication, such as an accessible Typst version of it, and attach that to your dissertation instead. However, also note that you should still refer to the peer-reviewed versions of your publications in the bibliography and on the attached publication title pages, such as is done in this template for the attached example publications (if the selected thesis type is “high” enough.)

[1] Including publications is not actually required for compilation PhD dissertations. Only referring to the publications via full bibliographic information is actually required.

7 REFERENCES

Different referencing styles determine how you create in-text citations and the bibliography (list of references). Two common referencing styles are presented in this chapter:

- Numeric referencing (Vancouver system), such as [1], [2], …
- Name-year (Harvard) system, such as (Weber 2001), (Kaunisto 2003), …

A numeric reference is inserted in square brackets, whereas the last name of the author and the year of publication are given in parentheses. Both styles are acceptable, but the conventions for referencing vary between disciplines. You must pick one and use it consistently throughout your thesis. The list of possible styles can be found on the Typst documentation page [50].

The most commonly used tool for creating bibliographies in Typst is the Hayagriva [51] YAML [52] format, but Typst also supports BibLaTeX [53]. The former is what this thesis template uses by default. Both Hayagriva and BibLaTeX are based on gathering bibliographic information from the sources to a bibliography file using a specialised syntax. Typst reads both this file and the document being written, and then forms the references and the bibliography based on this information. A citation style can be chosen by setting it in the `metadata.typ` file, in the variable `citationStyle`.

7.1 IN-TEXT CITATIONS

In-text citations are placed within the body of the text as close to the actual claim that the citation supports as possible. The citation is generally placed within the sentence before the next period:

- Weber argues that… [1].
- Cattaneo et al. introduce in their study [2] a new…
- The result isldots [1, p. 23]. One must also note… [1, pp. 33–36]
- In accordance with the presented theory… (Weber 2001).
- It must especially be noted… (Cattaneo et al. 2004).
- Weber (2001, p. 230) has stated …
- Based on literature in the field [1, 3, 5] …
- Based on literature in the field [1][3][5] …
- The topic has been widely studied [6–18] …
- … existing literature (Weber 2001; Kaunisto 2003; Cattaneo et al. 2004) has …

You *can* also place a citation after a period, in which case it refers to the entire preceding paragraph. However, this is generally not recommended, as it might make it unclear which claim the citation supports, if there are manu within the preceding paragraph.

7.2 THE BIBLIOGRAPHY FILE

Typst works similarly to LaTeX when it comes to citations and the bibliography:

1. A user collects a database of references into a bibliography file, where each reference has a *unique* identifier string that can be referenced, such as `mainauthor-etal-topic-year`.
2. The file is imported into a project using the `bibliography` function [50] of Typst. This is done in the file `main.typ` in the case of this template.
3. To cite a bibliography file entry with the identifier `id`, one writes `@id` within the text of their Typst project. As an example of what this might look like, placing `@Bezanson2017Julia` in a paragraph results in the following citation: [29].

The bibliogprahy file formats that Typst supports are *Hayagriva* [51] and *BibLaTeX* [53]. This template contains mostly equivalent examples of both respective formats in the form of the files `bibliography.yaml` and `bibliography.bib`.

The preferred format for this template is Hayagriva, the reason for which will be explained in Section 7.3. If you have an existing bibliography file in the form of BibLaTeX, Hayagriva can also be installed as a command line program [51, Installation], and the BibLaTeX file converted to Hayagriva format using the command

```
hayagriva bibliography.bib > bibliography.yaml
```

There are different kinds of citation types such as articles or web pages, which should be typeset differently in the bibliography. In the case of Hayagriva, an article reference metadata is given in Listing 7.1, whereas an equivalent BibLaTeX entry is shown in Listing 7.2.

```yaml
Bezanson2017Julia:
  type: article
  title: >-
    Julia: A fresh approach to numerical computing
  author:
  - Bezanson, Jeff
  - Edelman, Alan
  - Karpinski, Stefan
  - Shah, Viral B
  date: 2017
  page-range: 65-98
  serial-number:
    doi: 10.1137/141000671
  url:
    value: https://doi.org/10.1137/141000671
    date: 2025-10-10
  parent:
    type: periodical
    title: SIAM review
    publisher: SIAM
    issue: 1
    volume: 59
```

Listing 7.1. An example of a Hayagriva article entry.

```bibtex
@article{
  Bezanson2017Julia,
  title={{Julia: A fresh approach to numerical computing}},
  author={Bezanson, Jeff and Edelman, Alan and Karpinski, Stefan and Shah, Viral B},
  journal={SIAM review},
  volume={59},
  number={1},
  pages={65--98},
  year={2017},
  publisher={SIAM},
  url={https://doi.org/10.1137/141000671}
}
```

Listing 7.2. An example of a BibLaTeX article entry.

It is often preferable to order the list of references alphabetically by the first author's last name. This template however uses the `ieee` style by default, meaning that citations are listed in the bibliography in the order of appearance. Change the value of the variable `citationStyle` in the file `metadata.typ` to change this.

7.3 INSERTING PHD PUBLICATIONS INTO YOUR THESIS

As was mentioned in Section 6.7.3, the PDF format has some limitations regarding attaching PDF files into other PDF files, if one wishes for the final product to remain standard-conforming. Regardless, this template provides support for inserting PhD publications at the end of this document[2] using the bibliography file `bibliography.yaml`, if the export format is not one of the tagged variants such as PDF/A-3a or PDF/UA-1. You can therefore test how the PDF attachments would look like if you just compile your document without accessibility tags:

```
typst compile template/main.typ
```

To tell the template which of your references in `bibliography.yaml` should be considered as compilation dissertation attachments, you should add additional fields to the Hayagriva file `bibliography.yaml` entries, that correspond to your PhD publications. These are of the following form:

```yaml
path: publications/publication.pdf
license: null or the name of the license
n-of-pages: the number of pages in the document
tauthesis-publication: true or false
author-contribution: >
  A short description of how you contributed to this work.
  This can extend on multiple lines, as long as the lines
  are indented.
```

The `path` field should be in relation to the `main.typ` file. It might be a good idea to utilize the readily provided `publications/` folder in the template directory, where the example publications already reside.

7.4 ATTACHING TYPST-BASED PHD PUBLICATIONS

If you are in the lucky position of being allowed to use Typst source code to enter your dissertation publications into your thesis[3], it should be entirely possible to make your entire dissertation accessible. To aid with this, targeting *Open Access publishing channels* [48] and preferring licenses such as CC BY 4.0 [54] in your publications allows you to create a *derivative work* from the publication. A derivative work can be a Typst version of the publication.

The Typst source simply needs to adhere to related accessibility standard requirements [22], [32]. Simply point the `path` field of a Hayagriva bibliography entry to a valid Typst root file of your publication, and this template will `#include` it.

[2] Which is not compulsory for it to be accepted.
[3] Which you again do not *need* to do. It is just a bonus if you can.

Note 1: Any included Typst files should *not* set their own page size or margins in a way which overwrites the ones of this template, as the page sizes need to match. Remove any

```
#set page(...)
```

invocations from the start of the publication file, and let this template handle the page settings for you. Otherwise you are allowed to customize the appearance of the publications using `set` and `show` rules of Typst, unless you wish to use the style of this template. Any utilized fonts should be accessible, though, so keep that in mind.

Note 2: Typst does not support multiple bibliographies in version 0.14.0 natively, but it is a planned feature of version 0.15.0. In the meanwhile, you might wish to utilize a package such as `alexandria` or `pergamon` to typeset the bibliography of an attached Typst-based PhD publication and keep it separate from the bibliography of the main dissertation.

8 CONCLUSIONS

This template and the writing guidelines should help achieving consistently formatted and clear documents. Every writing and presentation must have a conclusion. This fact is here emphasized by having this short and rather artificial summary also in this template. The purpose of this chapter is to discuss how the things we set out to do in the introduction succeeded, and compare our results to other literature on the same topic.

As a last reminder, remember to make sure that your document is *accessible* [22] before submitting it to the Tampere University archive Trepo. See the university library guidelines [55] regarding this. Use veraPDF [46], showtags [43], PAC [44] (if you have access to Windows®) or some other software listed by the PDF Association [45] to verify the thesis PDF document accessibility, before your thesis submission.

If your thesis PDF file is not accessible and you have no way of making it so, you should attach both the unaccessible PDF file at least in PDF/A-2b format, and your *thesis source code* as a ZIP file to the Trepo submission. Be sure to clean up any personal comments from the source code before doing so. Also make sure that the final submitted source code works, meaning a PDF file can be compiled from it.

Note that you are allowed to attach the source code and a permissive license such as CC BY 4.0 [54] to your thesis, even if it is accessible. This is *the* ideal way of publishing your thesis, unless you are restricted from doing so by a company or other entity tied to your thesis process, that migth require secrecy from you. The authors of this template and the Tampere University Library recommend CC-licensing as the ideal licensing option [49], in the name of promoting open science.

9 TEST

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Appendix A: Example attachment

The point of attachments is to allow an author of a scientific work to attach research-related objects, that would interrupt the natural flow of the main matter or deviate from its main points, to the work. For example, large tables, long derivations of mathematical formulae or detailed implementations of algorithms in a specific programming language are better placed in the attachments.

In the main matter, it is better to present algorithmic descriptions in pseudo-code, as this allows the presentation to omit details related to the specific programming language being used, which makes the description more portable. In the case of mathematical theorems, the main matter presentation should contain at least the pre- and post-conditions of the theorem, but unless its proof is a main point of the work, it is better to put it into the attachments, and tell the reader which attachment it can be found from via a cross-reference.

Appendix B: Tests related to cross-referencing

This attachment is here just to make sure that cross-references work as intended. Towards this end, Figure B.1 is here just to make testing within the same chapter possible.

Figure B.1. A test image with a very long caption: Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo.

Also, here is an equation for similar purposes:

$$\mathbf{L}\mathbf{x} \approx \mathbf{y}\,. \tag{B.1}$$

Also, why not add a theorem to the mix.

Theorem B.1 (Pythagorean theorem). For sides a, b and c of a right triangle, where c is the *hypotenuse*, the following equality holds: $a^2 + b^2 = c^2$.

B.1 Figure tests

B.1 with an empty supplement as the first inline element of this paragraph, to make sure that there is no funny whitespace introduced before figure references.

Here is a reference to a figure in a previous chapter: Figure 6.1. Notice the period after the reference, meaning the spacing around the reference works as intended.

Here is a reference to the same figure but with an alternative supplement: Kuvassa 6.1.

Table reference: Table 6.1.

Table reference with an alternative supplement: Taulukossa 6.1.

Table reference with an empty supplement: 6.1.

B.2 Section tests

A chapter reference: 1.

A chapter reference with an alternative supplement: Luvussa 1.

1 as the first element of this paragraph to test that there is no unnecessary whitespace before heading references.

B.3 Equation tests

Here is an equation reference: $(B.1)$.

Reference equation in previous chapter: (6.1).

Check that supplements are ignored in the case of equation references: $(B.1)$.

$(B.1)$ as the first element of this paragraph. Again, no predecing whitespace should be generated.

B.4 Citation tests

A citation test: [29].

A citation test with page numbers: [29, s. 1–2]

[29]: no preceding whitespace at the start of a paragraph.

B.5 Theorem tests

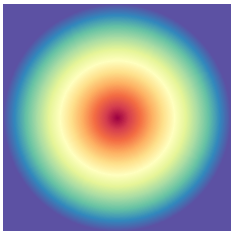
Referencing a theorem in this section: Theorem B.1.

Referencing a theorem box in another section: Example 6.1

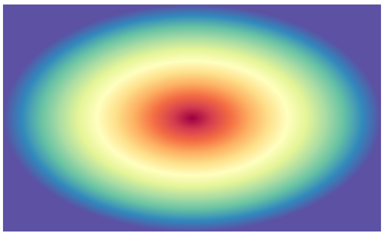
Referencing a theorem box in another section in Finnish: Esimerkki 6.2

B.6 Subfigure tests

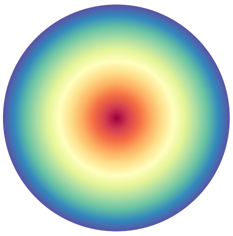
Here we test referencing subfigures in a different section. These are the linear gradient in Figure 6.2.a, the radial gradient in Figure 6.2.b and the conic gradient in Figure 6.2.c.



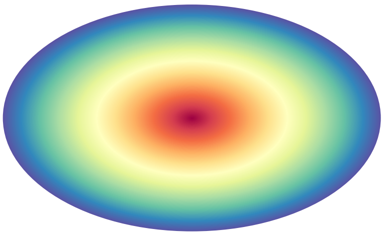
(a)



(b)



(c)



(d)

Shapes B.1. A few different kinds of shapes. Shapes B.1.a contains a square, Shapes B.1.b a rectangle, Shapes B.1.c a circle and Shapes B.1.d an ellipse.

Shapes B.1 is here for testing subfigures inside of figures of arbitrary kinds.

Appendix C: Test appendix with mathematics in its heading: $\mathbf{f} = m\mathbf{a}$. Also some `code`.

This test appendix is here to test how the text sizes in the lists of figures and such behave.

Figure C.1. A test figure with mathematics in the caption: $\int_{\{x\}} f(x)\,\mathrm{d}x$.

The text sizes should be sensible, and mathematics should also scale to the surrounding text size.

C.1 A level 2 heading with mathematics: $\int_{\{x\}} f(x)\,\mathrm{d}x$

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri.

C.1.1 A level 3 heading with mathematics: $\int_{\{x\}} f(x)\,\mathrm{d}x$

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri.

C.2 Another level 2 heading with mathematics: $\nabla \cdot \mathbf{j} = 0$ `more code`

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C.2.1 ANOTHER LEVEL 3 HEADING WITH MATHEMATICS: $\nabla \cdot \mathbf{j} = 0$

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C.3 A HEADING WITH `code`

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C.3.1 A HEADING WITH `more code`

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C.3.2 A HEADING WITH `even more code`

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